



**Sunbeam Institute of Information Technology**  
**Pune and Karad**

## **Operating System Concepts and Linux Programming**

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# File types and permissions

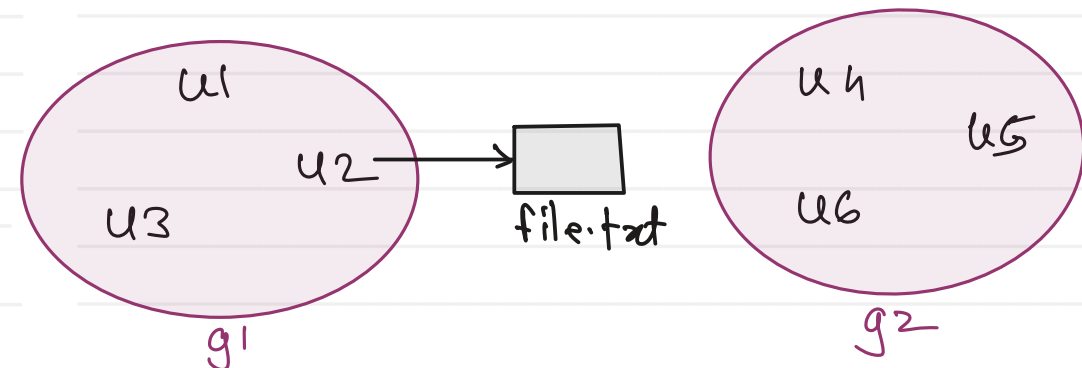
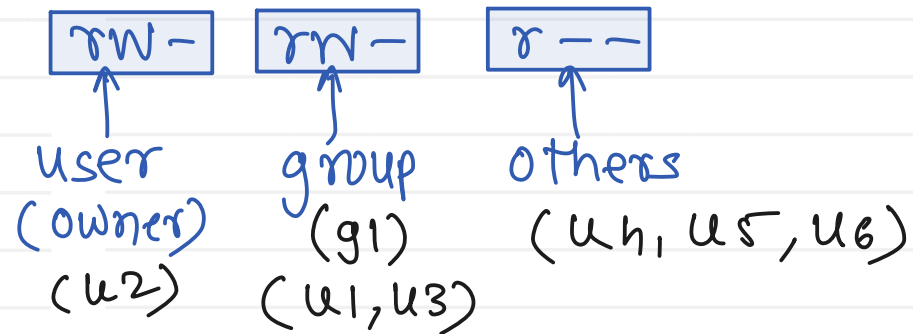
File permissions/mode/access:

File type: - rW-rW-r--

1. Regular file (-)
2. Directory file (d)
3. Link file (l)
4. Socket file (s)
5. Pipe file (P)
6. Char special file (c)
7. Block special file (b)

r - read  
w - write  
x - execute

Levels:  
user/owner  
group  
others



Human format :  $rW- rW- r--$   
Numeric format :  $\frac{110}{6} \frac{110}{6} \frac{100}{4}$   
: 0664 (octal)

Existing mode :  $rW- rW- r--$   
Add execute to user  
Remove write of group

New mode :  $rWX r-- r--$   
 $\frac{111}{7} \frac{100}{4} \frac{100}{4}$   
(0744)

## Types of users :

1. Super user (administrator)

username : root

Uid : 0

full privilege

- super user privilege can be taken by using sudo before commands

2. System users

users : mysql, apache, sshd

Uid : 1 - 999

- system users are no login users  
- limited privilege to a specific directory

3. Normal/human users :

Username : sunbeam

Uid : 1000 onwards

- with these users can do login  
- limited privilege to their home directory

file - file.txt

permissions = rw- rw- r--

user/owner = rw-

group = rw-

others = r--

chmod  $\pm$  r/w/x filename

+ - add

- - remove

chmod +x file.txt

chmod -x file.txt

chmod u/g/o  $\pm$  r/w/x filename

u - user/owner

g - group

o - others

chmod u+w file.txt

chmod g+r file.txt

chmod u+rwx file.txt

0 - 000

1 - 001

2 - 010

3 - 011

4 - 100

5 - 101

6 - 110

7 - 111

rw-

110

6

rw-

110

6

r--

100

4

(664)

rw-

111

7

rw-

100

4

r--

100

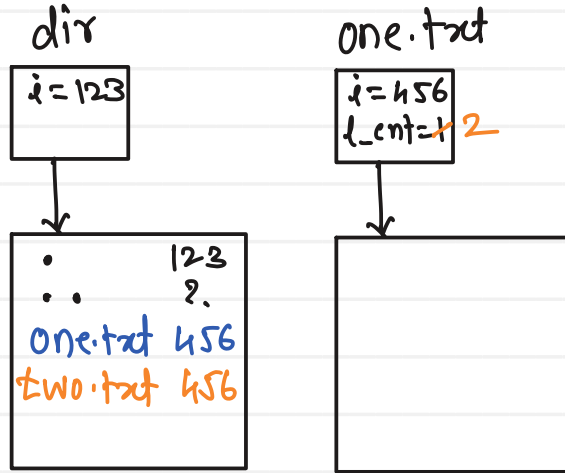
4

(744)

chmod mode filename

# Hard link and Symbolic link

In one.txt two.txt

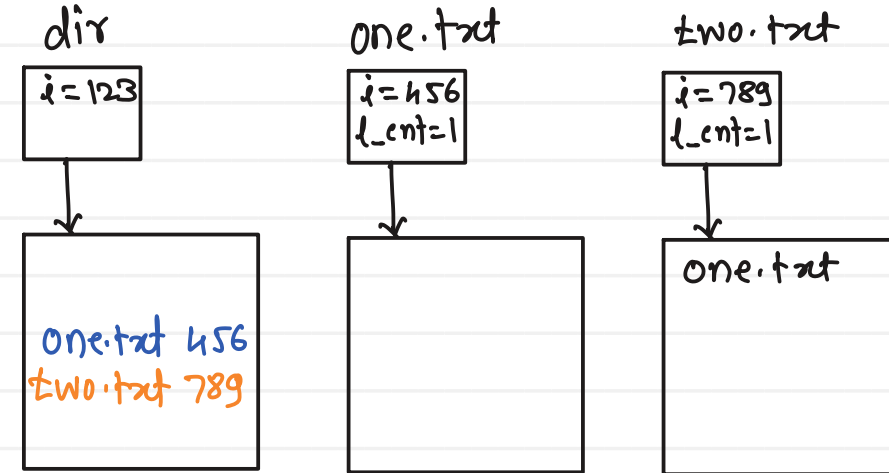


- hard links are tightly coupled with target files
- user can not create hard link of directories because it may lead to infinite loop while accessing the directories
- hard links can not be created across file systems

Applications:

- to rename a file
- space efficient backups
- File level deduplication
- Robust file access

In -s one.txt two.txt



- symbolic links are loosely coupled with target files
- symbolic links can be created for directories also
- symbolic links can be created across file systems

Applications:

- to create shortcuts
- software versioning & control
- cross file system references
- Path simplification
- Directory linking